

BASEL-MULHOUSE-FREIBURG AP, France

WMO: 072990

Lat: 47.6144N

Lon: 7.5100E

Elev: 885

StdP: 14.23

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 16.5	(c) 21.2	(d) 11.5	(e) 10.2	(f) 19.2	(g) 16.3	(h) 12.9	(i) 24.1	(j) 27.2	(k) 51.3	(l) 23.4	(m) 47.9	(n) 3.9	(o) 330	(p) 0.493

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 21.3	(c) 89.8	(d) 69.6	(e) 86.1	(f) 68.4	(g) 82.6	(h) 67.1	(i) 71.3	(j) 85.3	(k) 69.8	(l) 82.8	(m) 68.3	(n) 80.3	(o) 5.9	(p) 280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 66.8	(b) 101.7	(c) 76.2	(d) 65.4	(e) 96.9	(f) 74.4	(g) 64.2	(h) 92.8	(i) 72.9	(j) 35.6	(k) 85.7	(l) 34.2	(m) 82.9	(n) 33.0	(o) 80.5	(p) 78.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 18.9	(b) 16.1	(c) 13.7	(e) 9.7	(f) 95.4	(g) 7.0	(h) 3.5	(i) 4.7	(j) 97.9	(k) 0.6	(l) 99.9	(m) -3.3	(n) 101.9	(o) -8.4	(p) 104.4		
(a) 18.9	(b) 16.1	(c) 13.7	(e) 9.0	(f) 75.0	(g) 6.9	(h) 1.7	(i) 4.1	(j) 76.2	(k) 0.1	(l) 77.2	(m) -3.8	(n) 78.1	(o) -8.8	(p) 79.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.8	35.9	38.0	44.3	50.8	58.6	65.1	68.2	67.5	59.9	52.7	43.1	37.3		
	DBStd	13.36	7.88	7.92	6.92	6.76	6.31	6.45	5.74	5.88	5.72	6.78	6.81	7.47		
	HDD50	1729	439	339	199	72	7	0	0	0	2	52	222	397		
	HDD65	5169	902	757	641	426	217	78	33	40	173	384	658	860		
	CDD50	2405	2	2	23	97	272	454	564	543	298	134	14	2		
	CDD65	367	0	0	0	0	17	82	131	117	19	1	0	0		
	CDH74	3956	0	0	0	39	250	883	1388	1174	207	15	0	0		
	CDH80	1343	0	0	0	2	48	290	518	446	38	1	0	0		
(14) Wind	WSAvg	5.8	6.6	6.6	6.6	6.1	6.0	5.5	5.3	4.8	4.9	5.1	5.6	6.4		
(15) Precipitation	PrecAvg	30.5	1.9	2.0	2.0	2.4	3.1	3.5	2.9	4.1	2.1	1.9	2.8	1.8		
	PrecMax	42.1	3.7	6.0	4.0	4.6	5.1	7.2	5.3	7.1	6.0	4.2	7.0	3.3		
	PrecMin	20.4	0.2	0.4	0.4	0.6	1.6	0.4	1.7	1.1	0.4	0.5	0.4	0.2		
	PrecStd	5.2	0.9	1.3	0.9	1.1	1.0	1.5	0.9	1.7	1.3	1.0	1.6	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.3	61.0	70.0	78.8	85.3	92.5	94.9	95.6	84.6	75.8	63.9	56.7		
		MCWB	51.2	49.7	53.8	59.1	66.0	70.4	70.7	70.1	66.4	62.3	55.6	51.0		
	2%	DB	53.9	56.6	65.5	73.4	80.3	87.2	89.8	89.2	79.8	70.9	59.1	53.7		
		MCWB	48.8	47.9	51.8	56.6	62.8	69.1	69.6	69.4	65.9	60.7	53.0	49.4		
	5%	DB	50.2	53.0	60.4	68.8	76.3	83.1	85.8	84.8	75.6	66.8	55.6	50.5		
		MCWB	46.0	46.5	50.4	55.1	61.3	67.7	68.4	68.2	63.7	59.0	51.0	46.9		
	10%	DB	46.7	49.3	56.4	64.7	72.0	79.1	81.9	80.7	71.5	63.0	52.6	47.7		
		MCWB	43.5	44.4	48.7	53.4	59.7	65.8	67.0	67.4	61.8	57.2	49.1	44.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.2	51.3	55.5	60.6	68.1	73.8	73.1	73.2	69.1	64.0	56.5	52.1		
		MCDB	56.4	58.5	65.7	74.4	81.0	86.8	88.7	87.1	80.0	72.2	61.8	55.5		
	2%	WB	49.0	49.0	53.1	58.1	65.3	70.6	71.0	71.0	67.0	61.8	53.8	49.6		
		MCDB	53.2	55.4	61.8	70.3	77.0	83.4	85.8	84.8	77.3	68.6	58.1	53.1		
	5%	WB	46.2	46.9	51.2	56.1	62.8	68.8	69.4	69.4	64.8	59.8	51.6	47.4		
		MCDB	49.6	52.2	58.8	67.0	72.7	80.7	82.9	81.9	73.0	65.7	55.1	50.2		
	10%	WB	43.6	44.6	49.3	54.1	60.7	66.8	67.9	67.9	62.9	57.7	49.4	44.9		
		MCDB	46.6	49.1	56.1	63.2	69.7	77.2	79.6	79.3	70.2	62.5	52.3	47.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.8	14.0	18.0	20.2	19.5	20.5	21.3	21.0	19.8	16.8	12.3	10.3		
		MCDBR	14.8	20.8	26.1	28.8	28.0	27.8	29.0	28.9	26.7	22.3	17.0	13.2		
		MCWBR	10.6	13.0	14.9	14.6	13.1	12.5	11.6	12.0	13.7	12.7	11.2	9.8		
	5% WB	MCDBR	14.1	18.5	22.4	25.8	24.3	25.6	26.0	25.5	23.5	20.3	15.6	12.4		
		MCWBR	10.5	12.2	13.4	13.5	12.0	11.9	11.2	11.4	12.4	12.1	10.8	9.5		
(40) Clear-Sky Solar Irradiance	taub		0.311	0.336	0.370	0.407	0.416	0.421	0.420	0.413	0.390	0.367	0.332	0.308		
	taud		2.444	2.368	2.308	2.225	2.246	2.274	2.271	2.304	2.348	2.396	2.445	2.446		
	Ebn at Noon		245	260	268	268	269	268	266	263	258	245	233	230		
	Edh at Noon		21	28	35	41	42	41	41	38	33	27	21	19		
(44) All-Sky Solar Radiation	RadAvg		335	590	983	1375	1603	1888	1796	1560	1181	735	386	270		
	RadStd		46	96	153	205	162	172	162	149	115	67	46	53		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (112 neighbors)	N/A	N/A	N/A	+1.35	+0.70	+0.52	N/A	N/A	+94	+25	(47)

Nomenclature: See separate page